

Extracting Worker Unsafe Behaviors from Construction Images Using Image Captioning with Deep Learning–Based Attention Mechanism

Peichen Zhai¹; Junjie Wang, Ph.D.²; and Lite Zhang³

Abstract: Safety in the construction industry has always been a focus of attention. Existing methods of detecting unsafe behavior of workers relied primarily on manual detection. Not only did it consume significant time and money, but it also inevitably produced omissions. Currently, automated techniques for detecting unsafe behaviors rely only on the unsafe factors of workers' ontology to judge their behaviors, making it difficult to understand unsafe behaviors in complex scenes. To address the presented problems, this study proposed a method to automatically extract workers' unsafe behaviors by combining information from complex scenes—an image captioning based on an attention mechanism. First, three different sets of image captioning models were constructed using convolutional neural network (CNN), which are widely used in AI. These models could extract key information from complex scenes was constructed. Then, two datasets dedicated to the construction domain were created for method validation. Finally, three sets of experiments were conducted by combining the datasets and the three different sets of models. The results showed that the method could detect the worker's job type and output the interaction behavior between the worker and the target (unsafe behavior) based on the environmental information in the construction images. We introduced environmental information into the determination of workers' unsafe behaviors for the first time and not only output the worker's job type but also determine the worker's behavior. This allows the model output to be better for ergonomic analysis. DOI: [10.1061/JCEMD4.COENG-12096](https://doi.org/10.1061/JCEMD4.COENG-12096). © 2022 American Society of Civil Engineers.

Practical Applications: This study developed an intelligent solution for determining whether a worker had unsafe behavior in complex scenarios using behavioral norms. The operator would not need to prepare the appropriate construction safety knowledge, such as whether to wear a helmet, whether to wear a safety belt, or whether to work at height, but simply input the target image into the model, and the model would combine the predefined behavioral norms, scene information, and other factors to determine what kind of behavior (or unsafe behavior) was contained in the image and output a simple description of the information. Descriptions could also be set as fixed templates for easy management, such as worker A wearing (not) a helmet, and these descriptions would play a key role in daily management and project summaries. Using this method, managers could use the relevant equipment to automate the acquisition of possible good behaviors or violations of anyone on site. It also enables efficient organization and recording, improving the efficiency of managers.

Author keywords: Image captioning; Construction scene; Dataset; Deep learning; Unsafe behavior.

Introduction

The construction industry plays an important role in the economic system of a country (Yu et al. 2021). According to the China Construction Industry Association, the value-added of China's construction industry would be 7.18% of the gross domestic product in 2020. But the safety and health aspects of the global construction industry did not turn out well. In terms of safety alone, the construction industry was seen as one of the most dangerous areas of work (Poh et al. 2018; Ayhan and Tokdemir 2019), with one in five

workers dying in construction. And China is the world's largest construction market, a position it is expected to maintain shortly. The Chinese construction industry is characterized by a large number of companies and a large volume (Xu and Ding 2017). The safety issues arising from such a large scale would not be ignored.

Construction safety problems often arose for a variety of reasons, on the one hand, due to workers' unsafe behavior (Han et al. 2013; Mohajeri et al. 2021). On the other hand, the worker's surrounding environment, including tools, materials, and machinery, also has some influence on the worker's safety. With the development of computer vision technology, target detection (Hou et al. 2020; Wang et al. 2021) and image segmentation (Ali et al. 2022; Rahman et al. 2021) had been widely used in the research of construction safety. For example, Nath et al. (2020) used an improved algorithm of YOLO to achieve detection of whether workers on site are wearing PPE. Pi et al. (2021) implemented a semantic segmentation application (based on the Mask RCNN algorithm) on drones to segment key objects in images for preliminary assessment of postdisaster damage. Kong et al. (2021) used a segmentation algorithm to identify hazard areas and combined it with social-long and short-term memory (LSTM) to predict unsafe behavior of workers approaching the edge of the hazard area. The current research demonstrated that target detection and image segmentation

¹Graduate Student, Dept. of Civil Engineering, Ocean Univ. of China, Qingdao 266100, China. ORCID: <https://orcid.org/0000-0003-4247-0951>. Email: zhaipeichen@stu.ouc.edu.cn

²Lecturer, Dept. of Civil Engineering, Ocean Univ. of China, Qingdao 266100, China (corresponding author). ORCID: <https://orcid.org/0000-0002-1122-3336>. Email: wjj@ouc.edu.cn

³Graduate Student, Dept. of Civil Engineering, Ocean Univ. of China, Qingdao 266100, China. ORCID: <https://orcid.org/0000-0002-3937-0290>. Email: zhanglite3122@stu.ouc.edu.cn

Note. This manuscript was submitted on November 28, 2021; approved on September 15, 2022; published online on November 23, 2022. Discussion period open until April 23, 2023; separate discussions must be submitted for individual papers. This paper is part of the *Journal of Construction Engineering and Management*, © ASCE, ISSN 0733-9364.

algorithms are significant for extracting and identifying information about the location of targets or determining their presence. Both algorithms place a strong emphasis on the recognition of specific target features, while more valuable information in the image, such as information about the worker's relationship to the tools he uses and the surrounding environment, was often ignored by the algorithms. Such algorithms were specific and mechanistic in their feature extraction of visual information, thus greatly reducing the contribution of visual information to the detection of worker unsafe behavior.

In contrast, deep learning-based image captioning can integrate the information contained in the image and output a description of the behavior (unsafe behavior) that combines worker and environmental factors. This leads to a better determination of unsafe behaviors and improves the utilization of image information. Image captioning is a combination of computer vision (CV) and natural language processing (NLP) (Sun et al. 2017), which builds a bridge between construction image information and human semantic understanding by using an encoder architecture with added attention mechanisms to extract key information from images and then generating text descriptions through the decoder architecture. Its ability to automatically recognize images and output descriptive text has great potential in addressing image recognition and retrieval from construction sites. However, the public dataset (Lin et al. 2014; Young et al. 2014) has only a few images of construction scenes, making it not suitable for construction scenes.

To address the aforementioned issues, the main objective of this study is to propose a method to automatically extract workers' unsafe behaviors by combining key information from complex scenes. This study constructs three sets of image captioning models based on the model (Xu et al. 2015) using the PyTorch framework and three high-performance CNN networks (Resnet101, Resnet152, and Densenet161). Second, two construction domain-specific datasets (An et al. 2021) containing information on worker behavior were developed for experimentation. Dataset 1 is a small construction-specific dataset containing mainly descriptions of workers' job states, and Dataset 2 is a large construction-specific dataset containing five job types. Finally, the performance of the three different models was evaluated by BLEU score, training duration, and accuracy metrics. The method identifies the type of worker's job and outputs the interaction behavior between the worker and the object (unsafe behavior) based on the objects in the building image. Thus, the computer could accurately output the presence of unsafe behaviors of workers based on the behavioral information we define. This study introduced environmental information to the identification of unsafe worker behavior and the output of textual information would better assist in ergonomic analysis (Ryu et al. 2020).

Background and Related Work

With the continuous development of CV technology and deep learning (Paneru and Jeelani 2021), fine-grained information in architectural scenes was gradually used in a variety of analyses and calculations. Fig. 1 summarizes the practical uses of existing CVs. Visual information at construction sites could be classified into object recognition, motion recognition, and activity recognition in terms of the way it was collected and used.

Object recognition had been an important research topic in the field of architecture in the past decades (Czerniawski and Leite 2020). Deep learning-based object recognition methods had been mainly used to detect objects in construction site images. Cheng and Wang (2018) developed a method to automatically detect sewer

pipe defects from CCTV images. The method used CNN and was trained on 3,000 professional images, which enabled the eventually accurate identification of cracked objects in the pipe images. Object recognition also allowed for unsafe behavior monitoring (Fang et al. 2018a), where workers not wearing required helmets (Ding et al. 2018) or irregular operation of equipment could be flagged. Tracking construction equipment (Xiao and Zhu 2018; Konstantinou and Brilakis 2018) (e.g., backhoe excavators, cars, trucks) was also a very interesting area of research in object recognition in construction. Semantic segmentation of images using color attributes, the method had been used to identify and label objects in images (Kim et al. 2016; Pour Rahimian et al. 2020). Object recognition, as an early application of CV, laid the foundation for the development of automation in the construction field.

Motion recognition included workers' posture recognition and equipment posture recognition (Lee and Han 2021). Worker posture recognition enabled the detection and assessment of worker health conditions and safety behaviors. For example, Yu et al. (2019) used 3D views to capture the worker's posture, combined with smart insoles to assess the worker's joint load, and finally calculated the worker's workload. In addition, Yan et al. (2017) developed ergonomic posture recognition technology to address construction hazard prevention by developing view-invariant features based on skeletal motion from a common camera. The motion recognition of equipment is mainly for the pose recognition of earthmoving machinery (Xu and Yoon 2019). The posture recognition of equipment could also be used to predict its potential collision and achieve a safer construction environment (Luo et al. 2020). The generation of motion recognition further enhances the CV's knowledge of the behavior of a single body. However, the detection of unsafe behaviors would be better facilitated if information about the state of the machine and the worker could be considered together.

Activity recognition (Luo et al. 2021) was defined as the interaction between humans and objects, which might include steel, concrete, blocks, bricks, and other types of objects in a construction scene. Nowadays, it was a reality to determine the behavior directly by the worker's movement (Yang et al. 2016; Luo et al. 2018a, b; Gong et al. 2011). Ding et al. (2018) developed a new hybrid deep learning model by integrating CNN and LSTM to automatically identify unsafe worker behaviors. Liu et al. (2020) through the combination of CNN and LSTM could infer the state and behavior of workers from the information in the images. However, past research approaches were too focused on human insecurity and ignored the influence of the external environment on human unsafe states.

All these methods could perform automatic recognition of targets, target-to-target relationships, but failed to extract deeper, more valuable information, such as information about tools and environment in the image. In contrast, image captioning that could extract more information from images is not currently used to determine unsafe behaviors in construction scenes. Therefore, this study used an image captioning model with an additional attention mechanism to extract important object information and worker behavior information in the environment to more comprehensively identify unsafe behaviors in the building scenes.

Research Methodology

The proposed framework for outputting textual descriptions of unsafe behaviors in this study was shown in Fig. 2. First, image information containing workers' behaviors and unsafe behaviors is collected. Second, the textual information describing the behavior

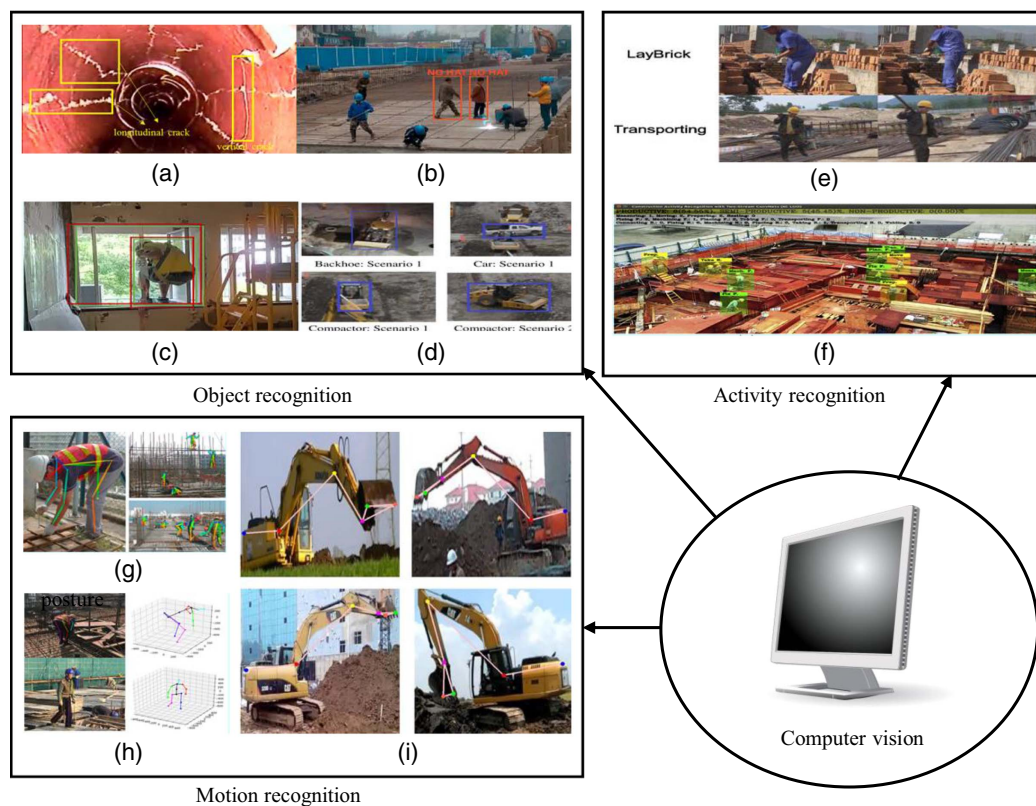


Fig. 1. Computer vision in the construction industry (a) pipe crack (reprinted from *Automation in Construction*, Vol. 95, J. C. P. Cheng and M. Wang, “Automated detection of sewer pipe defects in closed-circuit television images using deep learning techniques,” pp. 155–171, © 2018, with permission from Elsevier); (b) hardhat (reprinted from *Automation in Construction*, Vol. 85, Q. Fang, H. Li, X. Luo, L. Ding, H. Luo, T. M. Rose, and W. An, “Detecting non-hardhat-use by a deep learning method from far-field surveillance videos,” pp. 1–9, © 2018, with permission from Elsevier); (c) Object recognition also allowed for unsafe behavior monitoring (Fang et al. 2018a, b) worker (reprinted from *Automation in Construction*, Vol. 93, Q. Fang, H. Li, X. Luo, L. Ding, H. Luo, and C. Li, “Computer vision F1:6 aided inspection on falling prevention measures for steeplejacks in an aerial environment,” pp. 148–164, © 2018, with permission from Elsevier); (d) equipment (reprinted from Xiao and Zhu 2018, © ASCE); (e) single activity [reprinted from *Advanced Engineering Informatics*, Vol. 30(3), J. Yang, Z. Shi, and Z. Wu, “Vision-based action recognition of construction workers using dense trajectories,” pp. 327–336, © 2016, with permission from Elsevier]; (f) multiple activities (reprinted from *Automation in Construction*, Vol. 94, X. Luo, H. Li, D. Cao, Y. Yu, X. Yang, and T. Huang, “Towards efficient and objective work sampling: Recognizing workers’ activities in site surveillance videos with two-stream convolutional networks,” pp. 360–370, © 2018, with permission from Elsevier); (g) worker’s posture (reprinted from *Advanced Engineering Informatics*, Vol. 34, X. Yan, H. Li, C. Wang, J. O. Seo, H. Zhang, and H. Wang, “Development of ergonomic posture recognition technique based on 2D ordinary camera for construction hazard prevention through view-invariant features in 2D skeleton motion,” pp. 152–163, © 2017, with permission from Elsevier); (h) worker’s posture (reprinted from *Automation in Construction*, Vol. 94, H. Zhang, X. Yan, and H. Li, “Ergonomic posture recognition using 3D view-invariant features from single ordinary camera,” pp. 1–10, © 2018, with permission from Elsevier); and (i) excavator’s posture (reprinted from *Automation in Construction*, Vol. 110, H. Luo, M. Wang, P. K.-Y. Wong, and J. C. P. Cheng, “Full body pose estimation of construction equipment using computer vision and deep learning techniques,” 103016, © 2020, with permission from Elsevier).

is defined based on the relevant departmental specifications to produce a training dataset. Then, the feature extraction of the CNN network was enhanced by using the attention mechanism to focus on the features of tools, environment, and workers’ behaviors in the image information, and the correspondence between features and text was completed by LSTM. Finally, the text information was output so as to establish the connection between architectural images and human semantic understanding. In this section, the model application and dataset production were described.

Image Captioning Model

This section describes the image captioning of an attention mechanism-based encoding-decoding architecture, which is used to understand the image information of a complex construction site. Fig. 3 illustrated the structure of the image captioning model for

construction scenes. First, CNNs are used to construct encoders for visual feature extraction, such as region features, color features, and target features, from the construction images to capture deeper semantic information. Then, LSTM (Hochreiter and Schmidhuber 1997) networks are used to construct decoders to encode word positions and establish mapping relationships between image features and semantic features. In addition, an attention mechanism is added to make the model focus on features that better represent the environmental information to enhance the model’s perception and understanding of the architectural scene (Li et al. 2021).

Encoder

Encoders were used in CNNs to perform feature extraction on images. However, the complex and crowded environment of the construction site we were targeting places high demands on the feature extraction capabilities of the neural network. In this study,

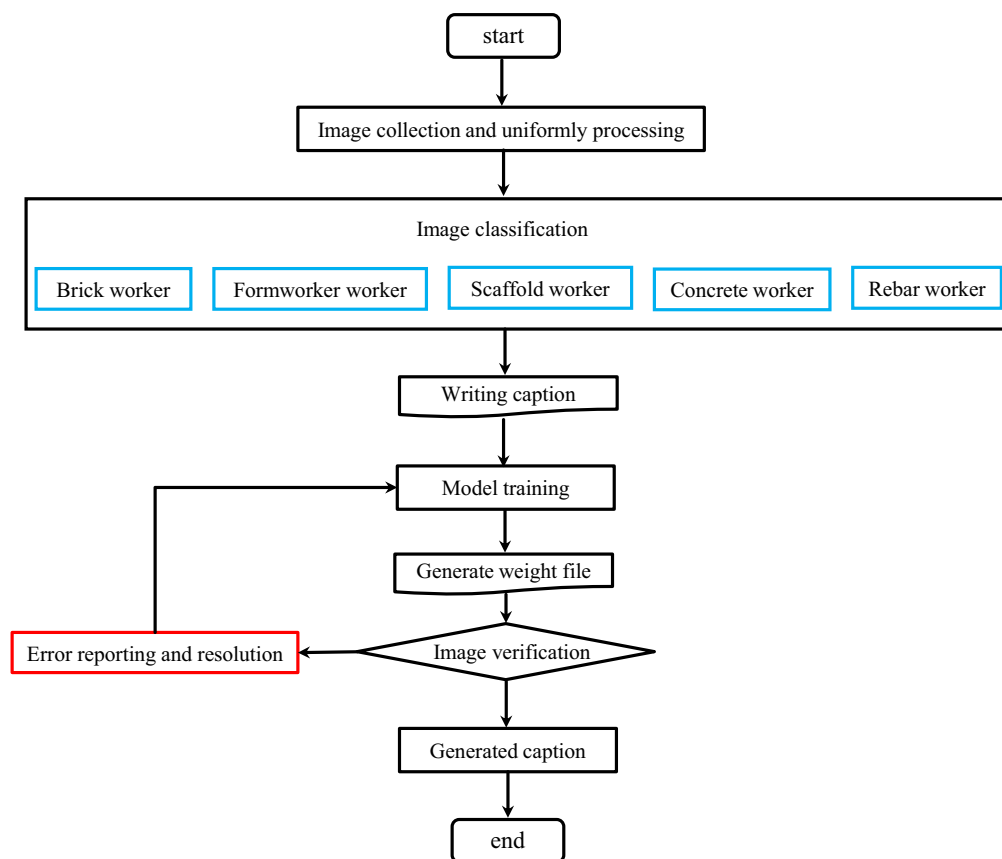


Fig. 2. Image captioning application process.

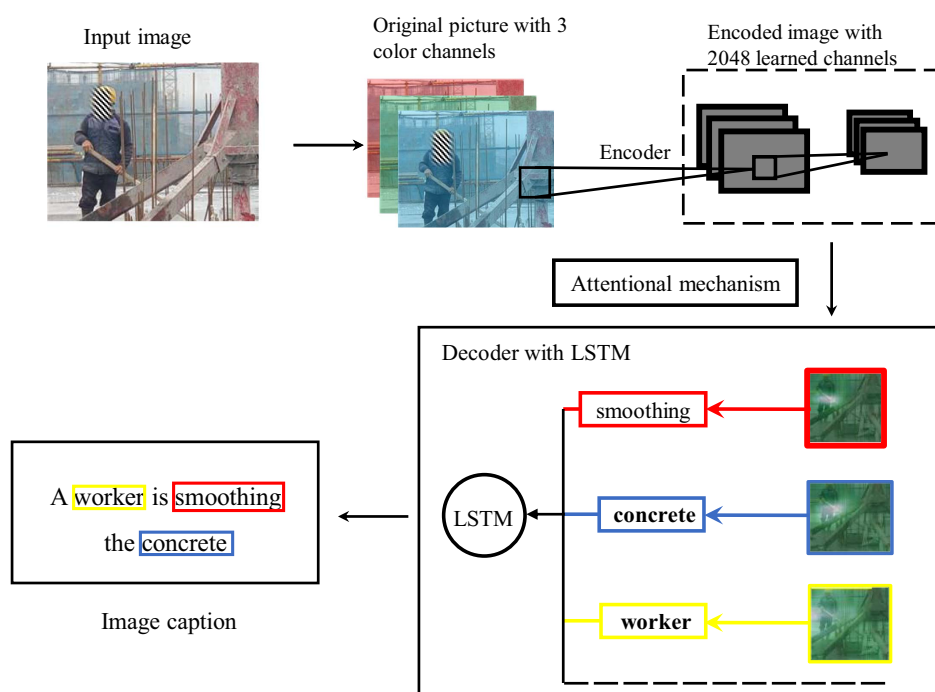


Fig. 3. Architecture diagram of image captioning with an attention mechanism. (Images by Peichen Zhai.)

the three CNN networks chosen, ResNet101, ResNet152, and DenseNet161, were all current state-of-the-art convolutional neural networks. The feature extraction ability and convergence speed in training were better than many neural networks such as VGG

(Simonyan and Zisserman 2014) and GoogLeNet (Szegedy et al. 2014).

ResNet101 and ResNet152 are both commonly used network structures in residual networks (He et al. 2016). ResNet152 has

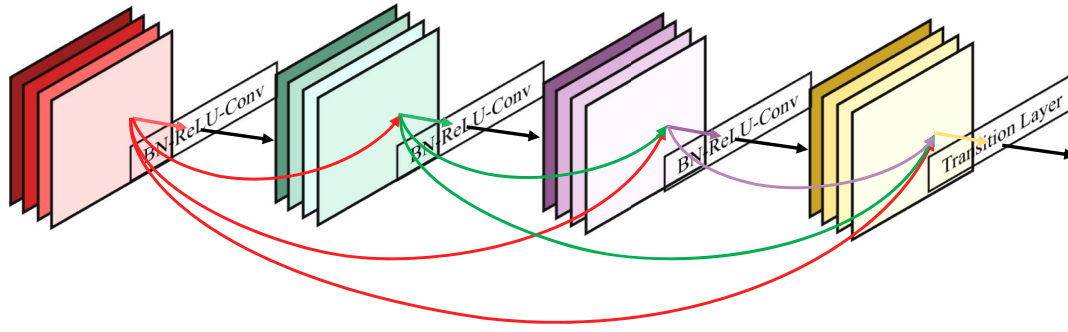


Fig. 4. DenseNet network feature propagation method.

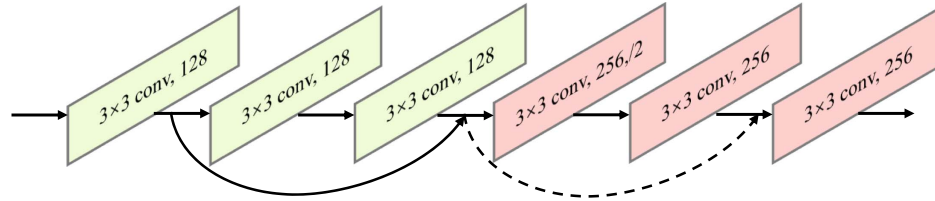


Fig. 5. ResNet network feature propagation method.

4 and 13 more convolutional layers in the conv2_x part and conv3_x part of the residual network than the ResNet101 network, respectively. Such a setup gives it eight more shortcuts than the ResNet101 network, which, to a certain extent, alleviates the possible problems of information loss and gradient explosion.

The essential difference between DenseNet (Huang et al. 2016) and ResNet is the way of feature propagation. As shown in Figs. 4 and 5, ResNet accepts only the feature information propagated from the previous layer, while DenseNet accepts the propagated features from each layer directly. Eqs. (1) and (2) explained the difference between ResNet and DenseNet, where X_l denotes the output of layer l and L -layer and H_l denotes a nonlinear transformation. As in Eq. (1), for ResNet, the input of layer l is the output of layer $l-1$ plus the nonlinear transformation of the output of layer $l-1$. As in Eq. (2) for DenseNet, the input of layer l is the nonlinear transformation that merges layer 0 to layer $l-1$. So ResNet has L -layer chained connections in a network with the same L -layer, but DenseNet would have $L(L+1)/2$ connections. This dense connectivity effectively alleviated the problem of gradient disappearance, enhanced feature propagation, and greatly reduced the number of parameters

$$X_l = H_l(X_{l-1}) + X_{l-1} \quad (1)$$

$$X_l = H_l([X_0, X_1, \dots, X_{l-1}]) \quad (2)$$

Decoder

In this paper, LSTM was used as the decoder of the model. LSTM was a kind of recurrent neural network after improvement, which could solve the problem of information loss in RNN in long sentences. In decoding, the most closely related words were selected according to the image features and their probabilities were calculated, and the word with the highest probability was selected as the output of the related image region. The first word of the output was marked as <start>, indicating the beginning of the whole subtitle output, and when the last word output was completed, it would be marked with <end> as the end marker. After the image features

were extracted, the model performed matrix operations on the current input word and word embedding matrix-generated feature vectors as the input of LSTM to obtain the probability values of different words and selected the word with the highest probability value as the output, whose calculation formula is shown in Eqs. (3) and (4)

$$\chi_t = W_e S_t, \quad t \in \{0 \dots N-1\} \quad (3)$$

$$P_{t+1} = \text{LSTM}(\chi_t), \quad t \in \{0 \dots N-1\} \quad (4)$$

To obtain the conditional probability of the output sequence S , the chain rule was used for the solution, as shown in Eq. (5)

$$P(S|I; \theta) = \prod_{t=0}^N P(S_t|S_0, S_1, \dots, S_{t-1}, I; \theta) \quad (5)$$

The LSTM encoding would use an implicit state h_t to store the initially generated words to memory up to the words generated at moment $t-1$. The new input x_t and the stored h_t are then updated by inputting a nonlinear function to obtain a new implicit state as in Eq. (6)

$$h_{t+1} = f(h_t, x_t) \quad (6)$$

Then, take the logarithm of Eq. (6) to obtain the log-likelihood function and use it as a loss function, as in Eq. (7)

$$\log P(S|I; \theta) = \sum_{t=0}^N \log P(S_t|S_0, S_1, \dots, S_{t-1}, I; \theta) \quad (7)$$

The model was trained by taking the sum of the log-likelihood functions of the maximized training samples as the objective, and the maximization probability of correctly described statements was obtained as shown in Eq. (8)

$$\theta^* = \arg \max_{\theta} \sum_{I, S} \log P(S|I; \theta) \quad (8)$$

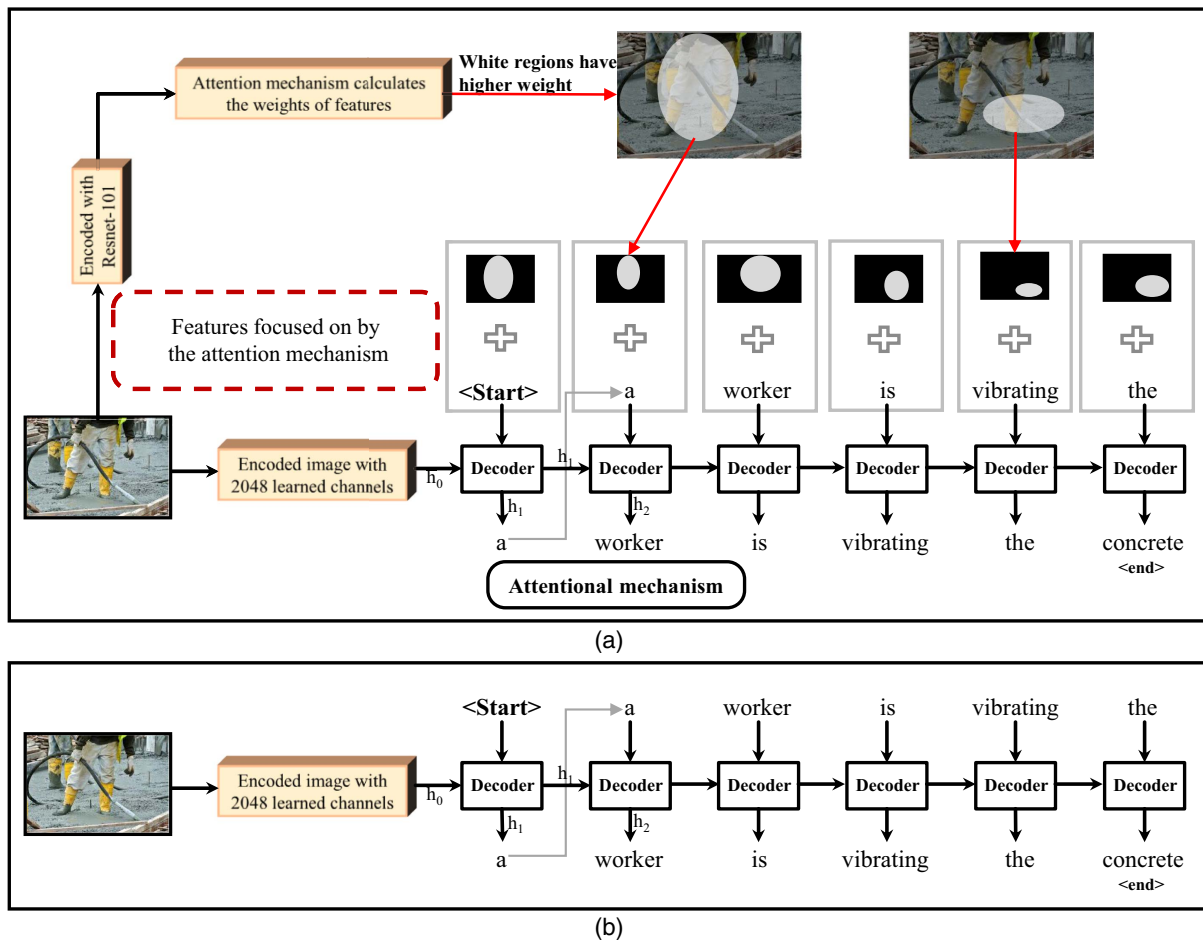


Fig. 6. Comparison of models with and without attention mechanisms (a) image captioning with attention mechanism; and (b) image captioning without attention mechanism. (Images by Peichen Zhai.)

Attentional Mechanism

Attention is a mechanism for assigning weight parameters (Vaswani et al. 2017). As shown in Fig. 6(a), the image captioning model with increased attention is easier than Fig. 6(b) to identify features with high salience and generate more accurate text descriptions based on that feature. In contrast, Fig. 6(b) would average the feature weights and subsequently performs text generation, which would tend to lead to information loss in the case of long sentences. Therefore, the introduction of an attention mechanism would allow the image captioning model to focus more attention on the focused regions and quickly capture the most needed information at the moment among a large amount of information.

The detailed flow of the attention mechanism is shown in Fig. 7, where h_t denotes the implicit state of the corresponding feature, with the word corresponding to the encoding state. The attention mechanism would calculate the attention weight, as shown in Eqs. (9) and (10), for them. The resulting attention weights could be used both in the form of a weighted summation of the content vector [Eq. (11)] and finally combined with h_t to obtain the output of the current decoding [Eq. (12)]

$$a_{t,s} = \frac{\exp(\text{score}(h_t, \bar{h}_s))}{\sum_{s'=1}^S \exp(\text{score}(h_t, \bar{h}_{s'}))} \quad (9)$$

$$\text{score}(h_t, \bar{h}_s) = \begin{cases} h_t^T \mathbf{W} \bar{h}_s \\ v^T \tanh(\omega_1 h_t + \omega_2 \bar{h}_s) \end{cases} \quad (10)$$

$$c_t = \sum_s a_{t,s} \bar{h}_s \quad (11)$$

$$a_t = f(c_t, h_t) = \tanh(\mathbf{W}_c [c_t; h_t]) \quad (12)$$

The visualization of the attention mechanism was shown in Fig. 8, where the highlighted area indicates the feature area that the model was currently focusing on. As we could see in Fig. 8(a), the attention mechanism filters the unimportant environmental features in this building scene and focused on the environmental information “rebar” so that the model perceived the presence of rebar.

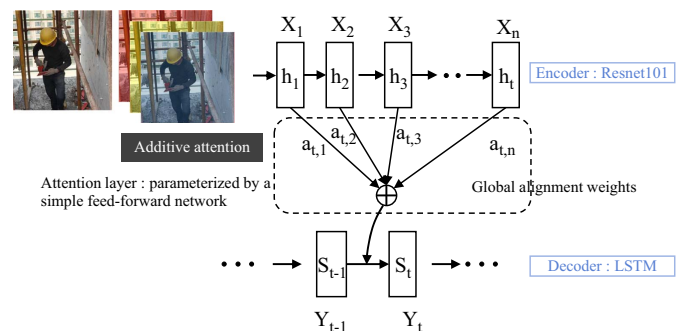


Fig. 7. Mechanisms of attentional mechanisms at work. (Image by Peichen Zhai.)

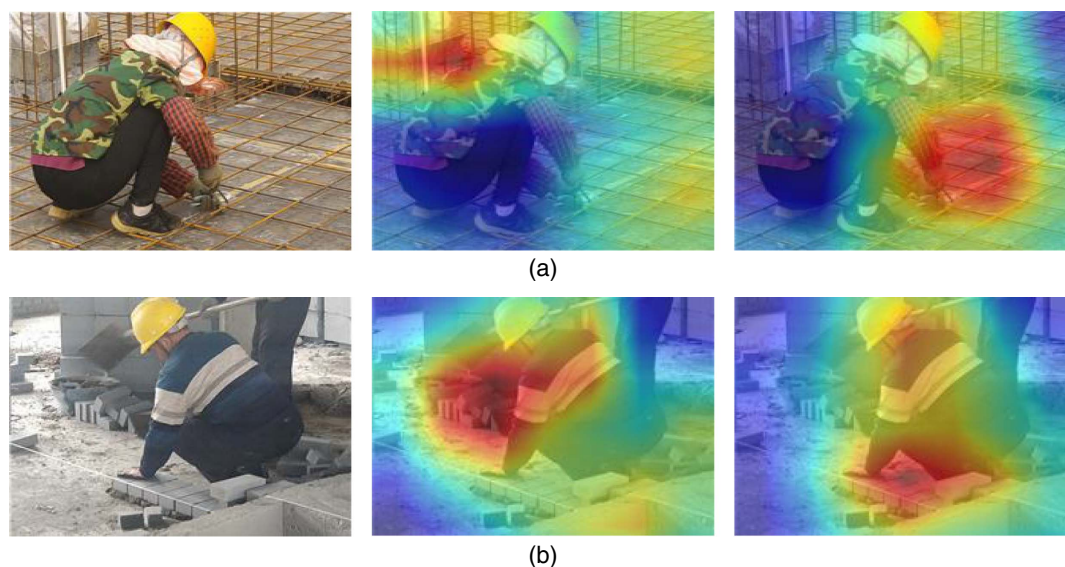


Fig. 8. Attention mechanism visualization effect (a) attention mechanism effect show 1; and (b) attention mechanism effect show 2. (Images by Peichen Zhai.)

Where attention was subsequently focused, other irrelevant environmental features were suppressed, allowing the model to correctly determine the worker's behavior. The scene in Fig. 8(b) shows a bricklayer laying bricks. Through feature filtering, the attention mechanism could correctly perceive and understand the "bricklayer" and "bricks" in the work content.

Production of the Datasets

In this study, two different datasets were created to verify the applicability of image captioning with attentional mechanisms in the construction domain. To ensure that the models were trained with image information from construction scenes, each image had to ensure at least a pair of subject-object relationships. The subject was the construction worker, and the object contained the objects that the worker was working or wearing (e.g., helmet, reflective undershirt, welding torch, rebar, concrete pipe, concrete, and formwork). Dataset 1 contained five worker behaviors in 659 images and contained descriptions of the relationships between workers and the protective tools they wore in their general state. Dataset 2 was produced centrally for worker unsafe behaviors. This dataset was relatively large and contained the behavioral actions of workers and their unsafe behaviors in the five major categories in 5,000 images and was used to validate the model's ability to perceive unsafe behaviors. The basis for the preparation of our two datasets is shown in Fig. 9, where boxes of the same color and linearity indicate the same elements. For example, the blue underlined indicated the object element. The dashed box at the bottom indicates the specific classification of each element. The details of the production of these two datasets are explained in the following sections.

Data Collection

Our two datasets were collected in two ways. For Dataset 1, we used keywords: construction workers, construction sites, and other keywords to crawl using spider crawler software. The downloaded images were first filtered to get the several scenes we had planned, and then the dataset was created. Dataset 2 was collected by taking live photos. To ensure that the dataset was realistic enough, we

collected them at three construction sites with different progress and different weather and lighting conditions.

Datasets Production Method

After classifying the collected images, the image size would be uniformized to 256×256 . In each image, it might contain multiple pairs of subject-object relations. To make full use of each image, each description in the generated sentences should contain a pair of relations, as shown in Fig. 10, where each description of the image was replaced with a different rule to describe all the contents in the image as much as possible. Each text description was generated manually, and then the words in the sentence were reorganized into "tokens" elements using a Python-related program or an Excel word-splitting program, e.g., "worker without helmet" was processed as "worker," "without," and "helmet." Then, the JSON file was generated for model training according to the MSCOCO dataset production standard.

Datasets Production Process

Based on the presented production method, in this section, we would introduce the production process of two kinds of datasets and the details of datasets production in different scenes.

The main purpose of making this Dataset 1 was to enable the model to perfectly identify the attributes of workers' dress and work status in different scenarios and to be able to describe the relationship between subject and object in one sentence with one worker as the subject. Therefore this dataset was collected in several main scenes, such as workers pouring concrete, workers welding steel, and workers carrying bricks. The specific figures in each scene are shown in Table 1. Each image, with at least one subject, also contains as many environmental objects as possible, such as the worker's hard hat and reflective undershirt. When manually generating image captions, two cases were encountered, one in which the subject was related to an object [e.g., the worker carrying bricks in Fig. 11(a)], and the other in which the subject was related to another object using a mediator [e.g., the worker pouring concrete with a pipe in hand in Fig. 11(b)]. If the first case was encountered, the connection between the subject and the object was generally with a verb, such as was moving and was pouring.

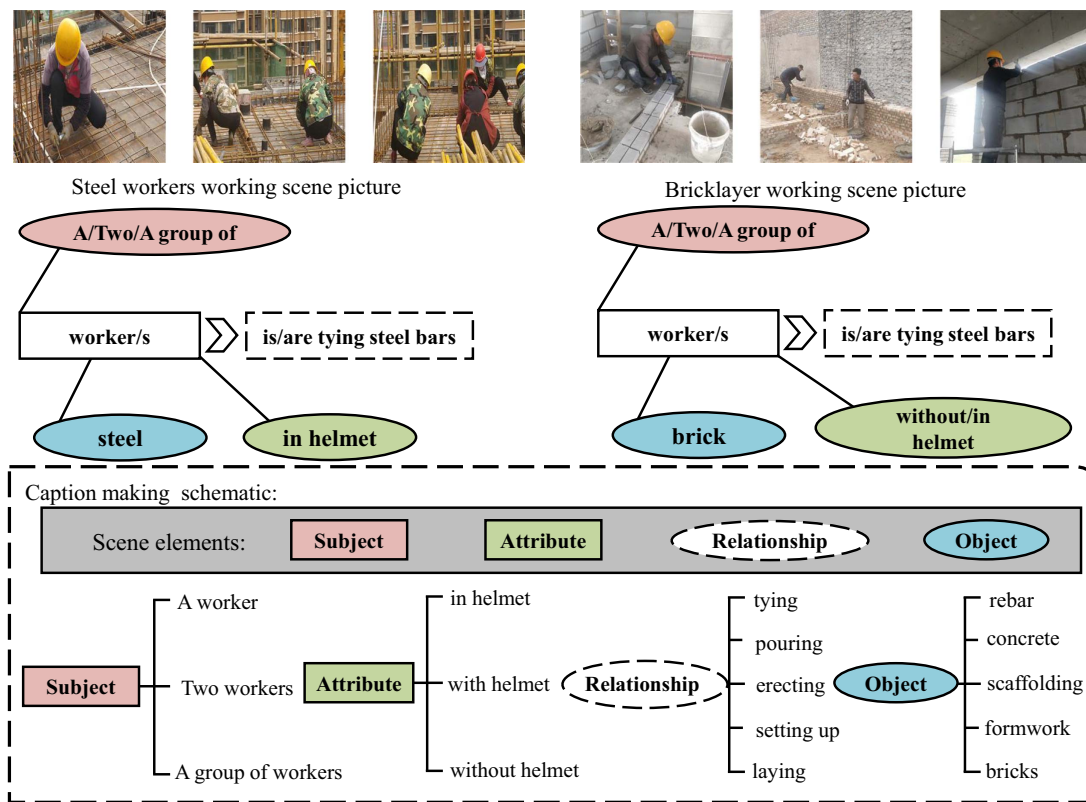


Fig. 9. Dataset production basis. (Images by Peichen Zhai.)

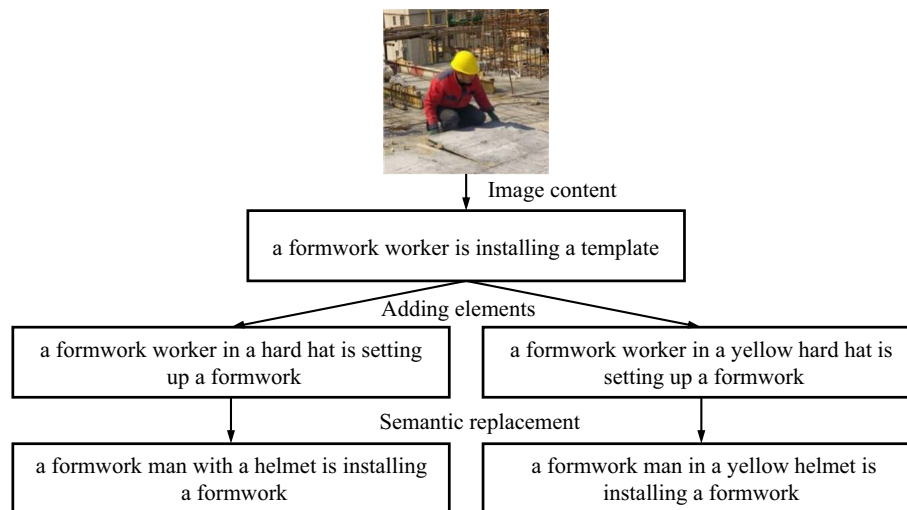


Fig. 10. Caption production basis. (Image by Peichen Zhai.)

The second case was generally described using the structural preparation of subject-medial-object.

Dataset 2 was produced mainly to allow the model to identify the worker's job type and work behavior by the surrounding environmental factors in the images and to test the stability of the model at the same time. The attribute categories that must be included in each image are shown in Fig. 12. The five types of worker information that we defined based on the external environment information were shown in Fig. 13 to describe their behaviors. For unsafe behaviors, based on the "General Specification for

Construction Scaffolding" (Ministry of Housing and Urban-Rural Development 2022): "scaffolding operations should be set up and dismantled with appropriate safety measures, operators should wear personal protective equipment and should wear non-slip shoes" and other specifications, we defined several cases: workers not wearing helmets, scaffolding workers (not) wearing personal protective equipment and other most typical unsafe behavior. Of course, the normative work behaviors account for most of the worker behaviors collected in the field, so if a worker was interacting with the rebar in the environment, we defined it as a rebar

worker was tying the rebar. If two workers were interacting with the concrete in the environment, we defined it as two concrete workers were pouring concrete. Like the one shown in Fig. 14, we defined the behavior of the workers in the five environments. We combined the elements extracted from the diagram into a sentence based on the production method described above to generate the caption description used for training.

Table 1. Specific number in each scene in Dataset 1

Scene category	Amount
Workers pouring concrete	91
Workers welding steel bars	140
Workers moving bricks	180
Workers weave reinforcing steel netting	126
Workers cutting rebar	122

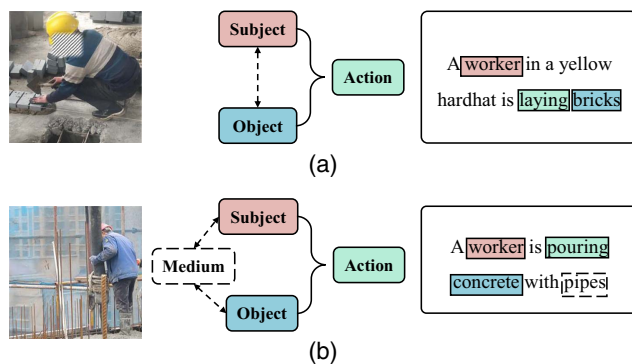


Fig. 11. Production method of Dataset (1) (a) Condition 1; and (b) Condition 2. (Images by Peichen Zhai.)

Experiments and Results

Data Collection and Annotation

The dataset was created as described previously. We wanted to make sure that each image had a person as the main object, clearly and unambiguously capturing the person at work. All images were then processed uniformly, with 75% of the images classified as the training set, 25% as the validation set, and 5% as the test set (included in the validation set). The datasets would be annotated using seven vocabularies such as nouns, verbs, and adjectives, and Fig. 15 counts the number of vocabularies used in our two datasets.

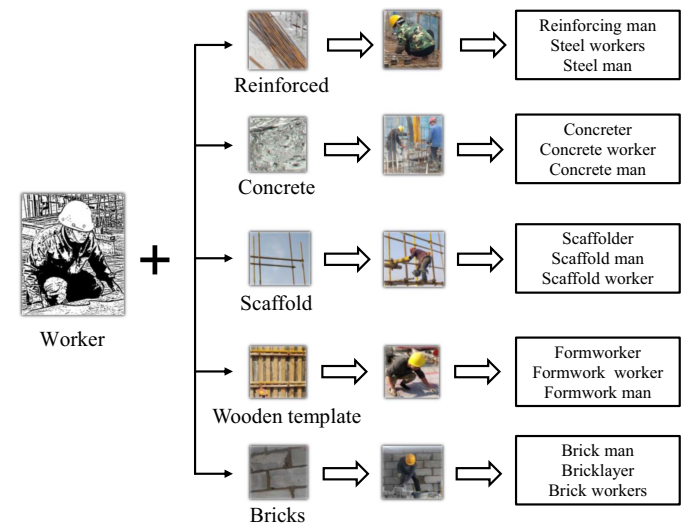


Fig. 13. Job classification standards for dataset 2. (Images by Peichen Zhai.)

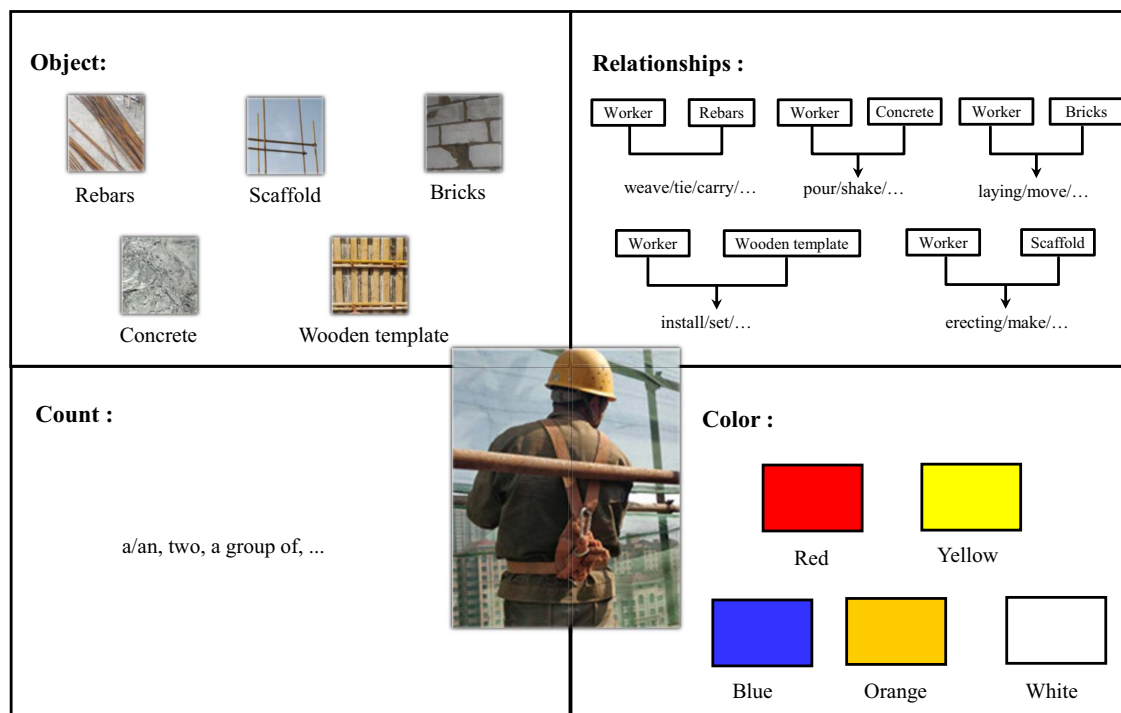


Fig. 12. Four attributes contained in a single image. (Images by Peichen Zhai.)

Rebar Worker		two steel workers are weaving steel bars two steel workers with hard hats are knitting steel bars two steel workers with helmets are weaving steel bars two steel men with hard hats are knitting steel bars two steel men with helmets are weaving steel bars
Concrete Worker		a concrete worker is casting concrete a concrete worker with a hard hat is pouring concrete a concrete worker in a yellow hard hat is pouring concrete a concrete worker is casting concrete with a pump pipe a concrete worker in a yellow helmet is pouring concrete
Scaffold Worker		a scaffold man was erecting the scaffolding a scaffold man with a hard hat was erecting the scaffold a scaffold man with a yellow hard hat was erecting the scaffold a scaffolder with a hard hat was erecting the scaffold a scaffolder with a yellow hard hat was erecting the scaffold
Formwork Worker		a formwork worker in a hard hat is setting up a formwork a formwork worker in a yellow hard hat is setting up a formwork a formwork man with a helmet is installing a formwork a formwork man in a yellow helmet is installing a formwork a formwork worker is install a template
Brick Worker		a bricklayer is laying bricks a bricklayer with a hard hat was laying bricks a brick worker with a hard hat was laying bricks a bricklayer in a yellow hard hat was laying bricks a tiler in a yellow hard hat was laying bricks

Fig. 14. Template for the production of dataset 2. (Images by Peichen Zhai.)

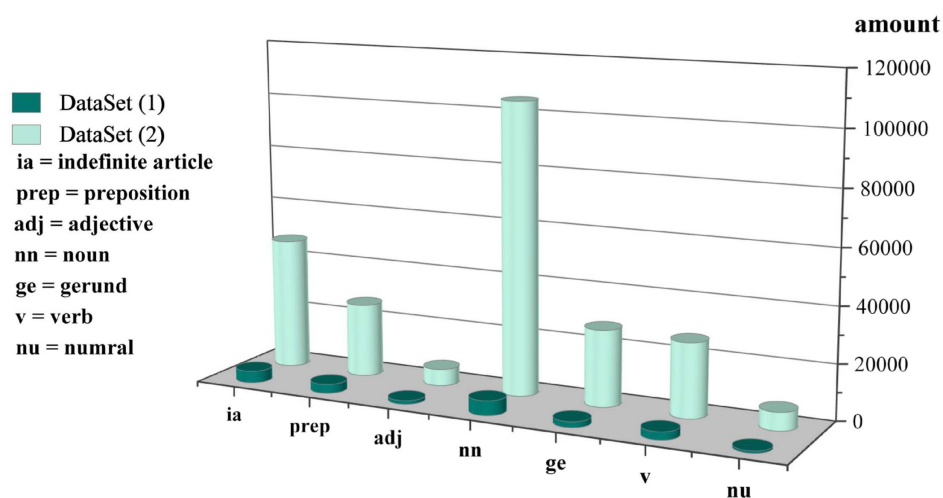


Fig. 15. Word usage statistics for the two datasets.

The classification criteria and the number of each in these two datasets respectively were counted in Tables 2 and 3.

Experimental Deployment

We performed three sets of simulated experiments, as shown in Table 4.

Three sets of simulation experiments investigated the impact of the size of the dataset and different CNN networks on the model performance. All three sets of experiments were conducted on a model developed based on PyTorch (Dutta et al. 2020), a python-based renewable computing package that uses a powerful

GPU to accelerate tensor computation. The experiments were conducted on an NVIDIA GeForce GTX1050 Ti. The setup of the experiments is shown in Table 5, with an initial learning rate of 4×10^{-4} . When the loss value was not converging in eight epochs, the learning rate was reduced to the original 0.8 to continue the training.

Experimental Results and Analysis

In our experiments, Precision, Recall, and BLEU values were used by us to assess the correctness of the generated captions. We recorded the BLEU scores measured under three sets of trials

(Papineni 2002). The BLEU value was the product of accuracy and phrase penalty factor as shown in Eq. (13), where ω_n was the current phrase that was occupied. From the formula, we could see that the value of BLEU takes the range of (0,1), so the higher the score, the better the machine translation. The “c” in the short sentence penalty factor (BP) Eq. (14) indicated the sentence length of the translation, and “r” indicated the sentence length of the reference translation. Precision (P_n) indicated the ratio of the number of relevant elements retrieved (TP) to the total number of retrieved scene elements (TP + FP), as shown in Eq. (15). The recall is indicated as the ratio of the number of relevant elements retrieved (TP) to the total number of relevant scene elements (TP + FN), as shown in Eq. (16)

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{i=1}^N \omega_n \cdot \log P_n\right) \quad (13)$$

$$\text{BP} = \begin{cases} 1 & c > r \\ e^{(1-\frac{c}{r})} & c \leq r \end{cases} \quad (14)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (15)$$

Table 2. Type statistics of Dataset 1

Classification	Train	Val	Combined
Workers poured the concrete	60	26	86
Workers welding steel bar	71	35	106
Workers move brick	120	50	170
Workers weave the steel skeleton	105	52	157
Workers cut steel bar	70	33	103

Table 3. Type statistics of Dataset 2

Classification	Train	Val	Combined
Rebar worker	795	340	1,135
Concrete worker	597	256	853
Scaffold worker	917	393	1,310
Formwork worker	624	267	891
Brick worker	710	307	1,017

Table 4. Deployment of three groups of experiments

Models	Datasets	Number
ResNet101+Attention+LSTM	Datasets 1	—
	Datasets 2	T1
ResNet152+Attention+LSTM	Datasets 2	T2
DenseNet161+Attention+LSTM	Datasets 2	T3

Table 5. Key parameter settings for the three sets of experiments

Major classes	Group	Batch size	Epoch
T1	1	32	30
	2	64	30
T2	1	32	30
	2	64	30
T3	1	32	30
	2	64	30

$$\text{Recall} = \frac{TP}{TP + FN} \quad (16)$$

In Test 1 (T1), the model was trained using Dataset 1 to output information on worker behavior in complex scenarios, but it no longer obtained reasonable scores for Beam sizes of 3 or 5. Nevertheless, it is still higher than the score measured in the MS COCO dataset (BLEU-4 of 0.369). To improve the accuracy of the model, we increased the number of images in Dataset 1 to achieve the desired score. Therefore, we set the number of images for each category in Dataset 2 to approximately one thousand. Three sets of experiments using Dataset 2 were conducted to validate the impact of CNN and batch_size on model performance, respectively. The obtained scores and training time are summarized in Table 6.

From Table 6, it could be concluded that T1 had the longest training time and the highest score at a batch_size of 32. T2 shortened the training time at a batch_size of 32, but the accuracy was slightly lower than T1. This indicates that the propagation of information in the ResNet network in the 152-layer transfer network resulted in partial loss of information, but its path helped to improve the speed of information propagation. T1 and T2 produced nonconvergence of loss values at a batch_size of 64 because the too-small dataset resulted in insufficient generalization of the model on the validation set. And there was a repeated calculation of the loss values during the training process, which made the training time longer than expected. T3 was a model built using the new DenseNet network. It could be seen that the training time was significantly shortened by the smaller number of parameters and the denser conduction network. The two sets of small experiments in T3 verify that a larger batch_size is beneficial to the convergence of the loss values and the improvement of the model accuracy under the appropriate model. However, compared with T1 and T2, the accuracy obtained by the T3 model was much lower and only higher than that obtained by training with the MS COCO dataset. Also, we compared the T1 scores with the generic models Adap (Lu et al. 2017), Self (Rennie et al. 2016), Updo (Anderson et al. 2017), Naby (Lu et al. 2018), and Mani (Liu et al. 2020), whose values were recorded in Table 7. To facilitate the comparison, the different

Table 6. Three sets of test scores

Major classes	Group	BLEU-1	BLEU-2	BLEU-3	BLEU-4	Training
T1	1	0.7483	0.6897	0.6543	0.6283	12 h 50 min
	2	0.7291	0.6652	0.6239	0.5915	11 h 53 min
T2	1	0.7456	0.6845	0.6466	0.6167	6 h 52 min
	2	0.7292	0.6591	0.6158	0.5839	12 h 2 min
T3	1	0.5694	0.4907	0.4462	0.4157	3 h 30 min
	2	0.6663	0.5875	0.5476	0.5194	2 h 43 min

Note: Bold indicates the best performing set of data for each group of experiments.

Table 7. Comparison of different model scores

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4
Self	0.742	0.576	0.435	0.327
Updo	0.759	0.595	0.455	0.337
Naby	0.762	0.602	0.456	0.342
Mani	0.68	0.595	0.537	0.491
Our	0.748	0.690	0.654	0.628

Note: Bold indicates the evaluation data of our experiments.

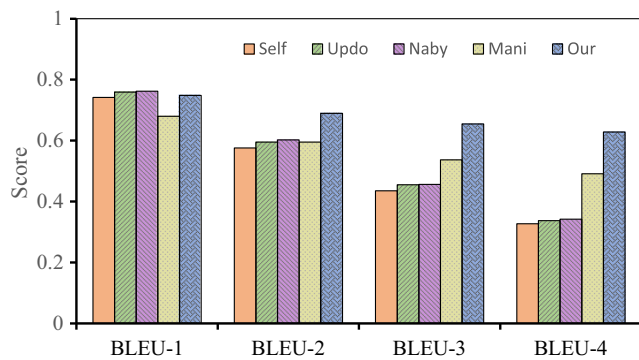


Fig. 16. Comparison of scores of different models.

model scores are given in Fig. 16. As shown in Fig. 16, the scores of the models with the addition of the attention mechanism all improved by more than 0.1 on average over the scores of the generic model, fully illustrating the superior performance of the proposed model in this study over the advanced CV methods.

Fig. 17 showed the detection process of the model for not wearing a helmet, and the highlighted area in the figure was the current attention mechanism focus area. First, it was obvious that the attention mechanism looked for relevant features in the attention region and output the words “without” and “brick” in combination with LSTM. Then, it accurately showed the unsafe behavior of the bricklayer who did not wear a helmet. With the help of these external environments, our judgment of unsafe behaviors would be further refined. To verify the validity of the model, we used

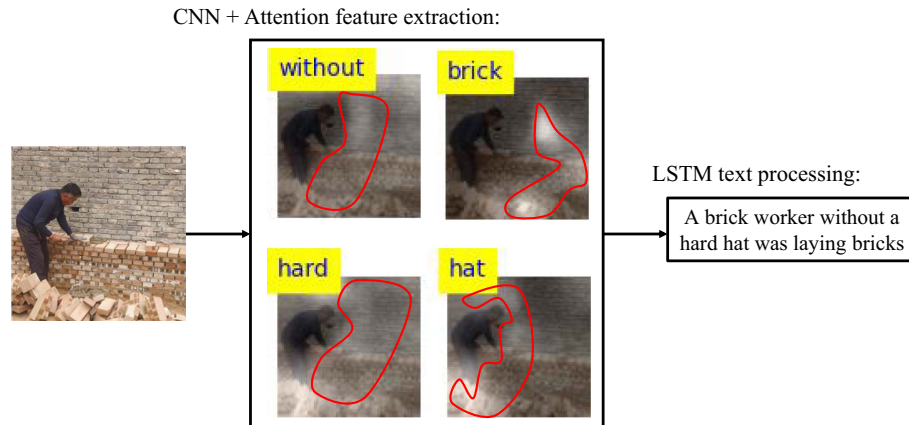


Fig. 17. Output process of the model. (Images by Peichen Zhai.)

Table 8. Accuracy and recall of different elements

Major categories	Minor categories	TP + FN	TP + FP	TP	P_n	Recall
Worker	Rebar worker	40	47	37	0.787	0.925
	Concrete worker	54	51	45	0.882	0.833
	Formwork worker	37	43	31	0.721	0.838
	Scaffold worker	38	36	35	0.972	0.921
	Brick worker	31	23	20	0.870	0.645
Color	Yellow	154	130	119	0.915	0.773
	Red	64	59	49	0.831	0.766
Target	Rebar	44	49	36	0.735	0.818
	Concrete	60	45	38	0.844	0.633
	Hard hat	214	167	142	0.850	0.664
	Steel pipes	27	23	20	0.870	0.741
	Scaffold	20	25	20	0.8	1
	Formwork	29	35	23	0.657	0.793
	Brick	18	21	16	0.762	0.889
Behavior	Tying rebar	35	40	29	0.725	0.829
	Vibrating concrete	41	30	27	0.9	0.659
	Holding steel pipe	13	12	12	1	0.923
	Erecting the scaffolding	36	46	30	0.652	0.833
	Laying bricks	20	31	18	0.581	0.9
	Pouring concrete	36	27	24	0.889	0.667
	Without helmet	61	65	56	0.862	0.918
	Installation template	45	40	28	0.7	0.622
	Not wearing seatbelt	20	22	17	0.773	0.85
Amount of workers	A	95	78	78	1	0.821
	Two	39	34	24	0.706	0.615
	A group of	66	88	66	0.75	1

200 images for testing under the T1 model. The statistics yielded an average output time of 1.5 s per image during the validation process, which indicated that the validation speed of the model could meet the efficiency requirements of the manager. In Table 8, we recorded several important objectives of the test, including the attributes of workers, colors, and targets in the images, and calculated the respective precision and recall rates. As observed in Table 8, the identification accuracy of unsafe behaviors all reached 0.7 or higher, while the accuracy of not wearing a helmet was more credible because of its larger sample. The high correct rates for the color attribute and the target attribute indicate that the final correct rate for the target was positively correlated with the adequacy of the corresponding sample. This was by the law of statistical large numbers, where the more adequate the underlying sample, the more accurate the learning effect.

In summary, the experimental results showed that the model in this study, with the help of the attention mechanism, accurately judged the worker's work type and unsafe behavior through

environmental information for the first time. Judging the work type and behavior by environmental information would better help in ergonomic analysis. Fig. 16 also showed the superior performance of our model with the added attention mechanism over the advanced CV approach. However, the three sets of tests also show that the dataset developed in this study could only perform well within a limited range (e.g., only at five worker types and small batch_size), so expansion of the dataset in terms of the types and number of workers it contains is necessary. Although our model performed well in manual validation, the real experimental results still had some error messages. As shown in Fig. 18, our model might generate correct descriptions as in Fig. 18(a) in the majority of cases, but some descriptions produce confusion in understanding individual features, as in Fig. 18(b), which misdescribed the working information. There were also cases like Fig. 18(c) where the number of features of workers was completely misunderstood and information about the workers' job types was misinterpreted.

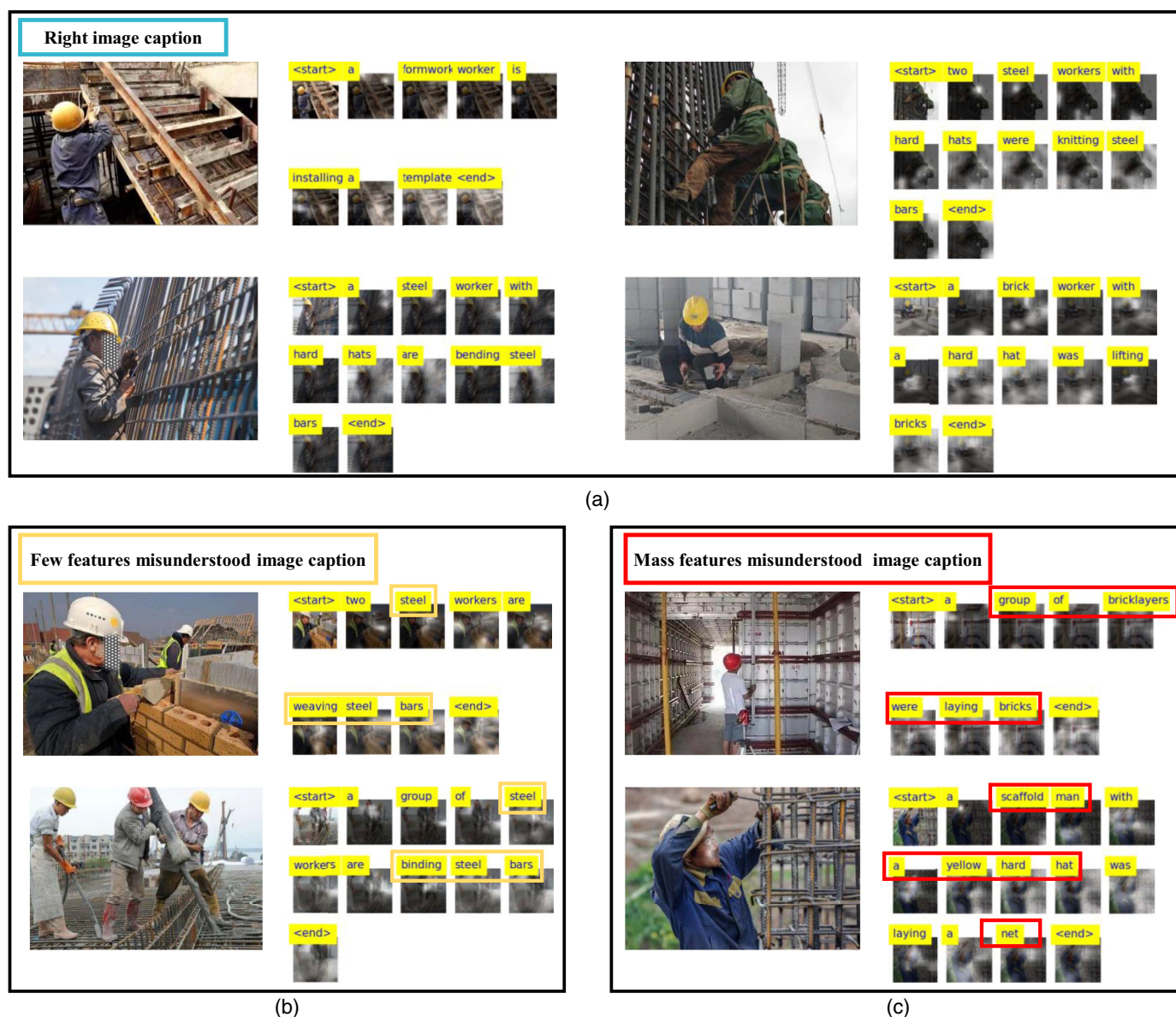


Fig. 18. Results of the model output: (a) right image caption; (b) few features misunderstood image caption; and (c) mass features misunderstood image caption. (Images by Peichen Zhai.)

Discussion and Future Work

This chapter would discuss the possibility of combining our produced datasets and image captioning in the context of building construction scenes and the focus of future work.

There are two main challenges to overcome in applying this approach in practice. The first is the extension of the dataset. As mentioned in the experimental results and analysis: the limitation of the dataset leads to the lack of generalization ability of the model. And there is no data collection for other unsafe behaviors such as workers standing beside cranes or workers working at heights without wearing seat belts. The second is to improve accuracy. The accuracy of the model is determined by the quantity and quality of the dataset and the intrinsic performance of the model. As described in the experimental results section, although our research model has high accuracy, there is still room for improvement.

Compared with current on-site unsafe behavior detection, the method in this study could achieve the detection of unsafe behavior at a low arithmetic cost. It was 13% of the image processing speed required for the safety warning of tower crane trajectory by Zhang and Ge (2022) and 18% of the instance segmentation required for the tracking implemented by Xiao et al. (2022), but the achieved accuracy was able to reach more than 80%. And this method outputs textual information that is understandable in nonconstruction specialized fields, which can realize zero-based applications in construction sites. This experiment used more advanced CNN networks (the three mentioned) than Liu et al. (2020) to improve the model, and it could be seen from the results that our test scores were all 10% higher. And the worker behavior (unsafe behavior) was further refined in the production of the dataset, such as distinguishing the behavior of five types of workers. However, the experimental results showed that although there was a reduction in the samples of each category after the differentiation of work types, the accuracy was still not affected.

Therefore, future work will focus on the exploration of practical applications of the model. First, we argue that target detection (Redmon et al. 2016) is complementary to image captioning. Target detection can play the role of detecting key targets from the video. If important safety gear such as helmets and safety undershirts do not appear in the video, they can be defined as key frames of unsafe behaviors so that keyframes can be extracted and inputted into our model to get detailed specific descriptions of unsafe behaviors. Our model can also make up for the fact that the target detection cannot output text information. Through the presented steps, we can achieve early warning and analysis of unsafe behavior. Second, there should be a large expansion of data from different construction environments, and collecting project-specific progress change data can help improve the generalization ability of the model. In addition, the unsafe behavior of human-machine interaction should also be emphasized, which can be achieved by introducing the identification of mechanical posture (Assadzadeh et al. 2022) to the identification of unsafe behavior. Worker tagging and sensors (Khan et al. 2022) can also be combined with the model at the same time. The resulting set of intelligent IoT systems can provide information about workers' activities in spatial and temporal dimensions. This is important for a comprehensive inference of workers' unsafe behaviors.

Conclusion

Environmental factors in building construction scenes are also a major cause of worker safety problems. Therefore, this study developed an image captioning that combines complex scene information to automatically extract worker job types and unsafe

behaviors. We applied image captioning for the first time in worker unsafe behavior detection. By introducing external environmental factors, not only the worker's work type was accurately determined, but also the output described the worker's behavior (unsafe behavior). The output of the model would better aid ergonomic analysis. Experimental results on a self-developed dataset showed that our model could use environmental factors as a basis for determining worker categories and worker behaviors, achieving a more comprehensive perception of behavioral information. In addition, the configuration required for the whole experiment is not expensive, requiring only one GPU (GTX1050 Ti or above) to implement, so the cost of deploying this model is low. The output of the model could provide not only unsafe behavior tips for professionals but also textual information on daily production for non-professionals, enabling a zero-based application in the field. However, there is still room for improvement, and the discussion of the results illustrates the need for more complete dedicated datasets for building construction scenarios and improved model performance. Exploring the application of the system will help to significantly increase the likelihood of project implementation. The code used in this study and the production dataset are open-source data and can be downloaded by the URL: <https://pan.baidu.com/s/1QQGY8m740srtDtGL6oeWUA>, extraction code: IPPQ.

Data Availability Statement

Some or all data, models, or codes that support the findings of this study are available from the corresponding author upon reasonable request. (The dataset and model code are available from the first author upon request.)

Acknowledgments

This study was supported by the key R&D program of Shandong Province. The authors are very grateful to all laboratory staff for their help and to Hongyu Chang for his online guidance.

References

- Ali, R., J. H. Chuah, M. S. Abu Talip, N. Mokhtar, and M. A. Shoaib. 2022. "Structural crack detection using deep convolutional neural networks." *Autom. Constr.* 133 (Jan): 103989. <https://doi.org/10.1016/j.autcon.2021.103989>.
- An, X. H., L. Zhou, Z. G. Liu, C. Z. Wang, P. F. Li, and Z. W. Li. 2021. "Dataset and benchmark for detecting moving objects in construction sites." *Autom. Constr.* 122 (Feb): 103482. <https://doi.org/10.1016/j.autcon.2020.103482>.
- Anderson, P., X. He, C. Buehler, D. Teney, M. Johnson, S. Gould, and L. Zhang. 2017. "Bottom-up and top-down attention for image captioning and visual question answering." In *Proc., IEEE Conf. on Computer Vision and Pattern Recognition*, 6077–6086. New York: IEEE.
- Assadzadeh, A., M. Arashpour, I. Brilakis, T. Ngo, and E. Konstantinou. 2022. "Vision-based excavator pose estimation using synthetically generated datasets with domain randomization." *Autom. Constr.* 134 (Feb): 104089. <https://doi.org/10.1016/j.autcon.2021.104089>.
- Ayhan, B. U., and O. B. Tokdemir. 2019. "Predicting the outcome of construction incidents." *Saf. Sci.* 113 (11): 91–104. <https://doi.org/10.1016/j.ssci.2018.11.001>.
- Cheng, J. C. P., and M. Z. Wang. 2018. "Automated detection of sewer pipe defects in closed-circuit television images using deep learning techniques." *Autom. Constr.* 95 (Nov): 155–171. <https://doi.org/10.1016/j.autcon.2018.08.006>.
- Czerniawski, T., and F. Leite. 2020. "Automated digital modeling of existing buildings: A review of visual object recognition methods."

- Autom. Constr.* 113 (Jun): 103131. <https://doi.org/10.1016/j.autcon.2020.103131>.
- Ding, L. Y., W. L. Fang, H. B. Luo, P. E. D. Love, B. T. Zhong, and X. Ouyang. 2018. "A deep hybrid learning model to detect unsafe behavior: Integrating convolution neural networks and long short-term memory." *Autom. Constr.* 86 (Jun): 118–124. <https://doi.org/10.1016/j.autcon.2017.11.002>.
- Dutta, S., A. Shi, R. Choudhary, Z. Zhang, A. Jain, and S. Misailovic. 2020. "Detecting flaky tests in probabilistic and machine learning applications." In *Proc., 29th ACM SIGSOFT Int. Symp. on Software Testing and Analysis*, 211–224. New York: Association for Computing Machinery. <https://doi.org/10.1145/3395363.3397366>.
- Fang, Q., H. Li, X. C. Luo, L. Y. Ding, H. B. Luo, and C. Q. Li. 2018a. "Computer vision aided inspection on falling prevention measures for steeplejacks in an aerial environment." *Autom. Constr.* 93 (Sep): 148–164. <https://doi.org/10.1016/j.autcon.2018.05.022>.
- Fang, Q., H. Li, X. C. Luo, L. Y. Ding, H. B. Luo, T. M. Rose, and W. P. An. 2018b. "Detecting non-hardhat-use by a deep learning method from far-field surveillance videos." *Autom. Constr.* 85 (Jan): 1–9. <https://doi.org/10.1016/j.autcon.2017.09.018>.
- Gong, J., C. H. Caldas, and C. Gordon. 2011. "Learning and classifying actions of construction workers and equipment using Bag-of-Video-Feature-Words and Bayesian network models." *Adv. Eng. Inf.* 25 (4): 771–782. <https://doi.org/10.1016/j.aei.2011.06.002>.
- Han, S., S. Lee, and F. Pena-Mora. 2013. "Vision-Based detection of unsafe actions of a construction worker: Case study of ladder climbing." *J. Comput. Civ. Eng.* 27 (6): 635–644. [https://doi.org/10.1061/\(ASCE\)Cp.1943-5487.0000279](https://doi.org/10.1061/(ASCE)Cp.1943-5487.0000279).
- He, K., X. Zhang, S. Ren, and J. Sun. 2016. "Deep residual learning for image recognition." In *Proc., 2016 IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*. New York: IEEE.
- Hochreiter, S., and J. Schmidhuber. 1997. "Long short-term memory." *Neural Comput.* 9 (8): 1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735>.
- Hou, X., Y. Zeng, and J. Xue. 2020. "Detecting structural components of building engineering based on deep-learning method." *J. Constr. Eng. Manage.* 146 (2): 04019097. [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0001751](https://doi.org/10.1061/(ASCE)CO.1943-7862.0001751).
- Huang, G., Z. Liu, V. Laurens, and K. Q. Weinberger. 2016. "Densely connected convolutional networks." In *Proc., IEEE Conf. on Computer Vision and Pattern Recognition*, 4700–4708. New York: IEEE.
- Khan, M., R. Khalid, S. Anjum, N. Khan, S. Cho, and C. Park. 2022. "Tag and IoT based safety hook monitoring for prevention of falls from height." *Autom. Constr.* 136 (Apr): 104153. <https://doi.org/10.1016/j.autcon.2022.104153>.
- Kim, H., K. Kim, and H. Kim. 2016. "Data-driven scene parsing method for recognizing construction site objects in the whole image." *Autom. Constr.* 71 (2): 271–282. <https://doi.org/10.1016/j.autcon.2016.08.018>.
- Kong, T., W. L. Fang, P. E. D. Love, H. B. Luo, S. J. Xu, and H. Li. 2021. "Computer vision and long short-term memory: Learning to predict unsafe behaviour in construction." *Adv. Eng. Inf.* 50 (Apr): 101400. <https://doi.org/10.1016/j.aei.2021.101400>.
- Konstantinou, E., and I. Brilakis. 2018. "Matching construction workers across views for automated 3D vision tracking on-site." *J. Constr. Eng. Manage.* 144 (7): 04018061. [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0001508](https://doi.org/10.1061/(ASCE)CO.1943-7862.0001508).
- Lee, K., and S. Han. 2021. "Convolutional neural network modeling strategy for fall-related motion recognition using acceleration features of a scaffolding structure." *Autom. Constr.* 130 (Jun): 103857. <https://doi.org/10.1016/j.autcon.2021.103857>.
- Li, T. S., M. Alipour, and D. K. Harris. 2021. "Mapping textual descriptions to condition ratings to assist bridge inspection and condition assessment using hierarchical attention." *Autom. Constr.* 129 (Apr): 103801. <https://doi.org/10.1016/j.autcon.2021.103801>.
- Lin, T.-Y., M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick. 2014. *Microsoft COCO: Common objects in context*. Berlin: Springer.
- Liu, H., G. B. Wang, T. Huang, P. He, M. Skitmore, and X. C. Luo. 2020. "Manifesting construction activity scenes via image captioning." *Autom. Constr.* 119 (5): 103334. <https://doi.org/10.1016/j.autcon.2020.103334>.
- Lu, J., C. Xiong, D. Parikh, and R. Socher. 2017. "Knowing when to look: Adaptive attention via A visual sentinel for image captioning." In *Proc., 30th IEEE Conf. on Computer Vision and Pattern Recognition (CVPR 2017)*, 3242–3250. New York: IEEE.
- Lu, J., J. Yang, D. Batra, and D. Parikh. 2018. "Neural baby talk." In *Proc., CVPR IEEE*, 7219–7228. New York: IEEE.
- Luo, H., M. Wang, P. K.-Y. Wong, and J. C. P. Cheng. 2020. "Full body pose estimation of construction equipment using computer vision and deep learning techniques." *Autom. Constr.* 110 (Feb): 103016. <https://doi.org/10.1016/j.autcon.2019.103016>.
- Luo, H., M. Wang, P. K.-Y. Wong, J. Tang, and J. C. P. Cheng. 2021. "Construction machine pose prediction considering historical motions and activity attributes using gated recurrent unit (GRU)." *Autom. Constr.* 121 (Jan): 103444. <https://doi.org/10.1016/j.autcon.2020.103444>.
- Luo, X., H. Li, D. Cao, F. Dai, J. Seo, and S. Lee. 2018a. "Recognizing diverse construction activities in site images via relevance networks of construction-related objects detected by convolutional neural networks." *J. Comput. Civ. Eng.* 32 (3): 04018012. [https://doi.org/10.1061/\(ASCE\)Cp.1943-5487.0000756](https://doi.org/10.1061/(ASCE)Cp.1943-5487.0000756).
- Luo, X. C., H. Li, D. P. Cao, Y. T. Yu, X. C. Yang, and T. Huang. 2018b. "Towards efficient and objective work sampling: Recognizing workers' activities in site surveillance videos with two-stream convolutional networks." *Autom. Constr.* 94 (Oct): 360–370. <https://doi.org/10.1016/j.autcon.2018.07.011>.
- Ministry of Housing and Urban-Rural Development. 2022. *General specifications for construction scaffolding*. GB 55023-2022. Beijing: Ministry of Housing and Urban-Rural Development.
- Mohajeri, M., A. Ardeshtir, H. Malekitabar, and S. Rowlinson. 2021. "Structural model of internal factors influencing the safety behavior of construction workers." *J. Constr. Eng. Manage.* 147 (11): 04021156. [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0002182](https://doi.org/10.1061/(ASCE)CO.1943-7862.0002182).
- Nath, N. D., A. H. Behzadan, and S. G. Paal. 2020. "Deep learning for site safety: Real-time detection of personal protective equipment." *Autom. Constr.* 112 (12): 103085. <https://doi.org/10.1016/j.autcon.2020.103085>.
- Paneru, S., and I. Jeelani. 2021. "Computer vision applications in construction: Current state, opportunities & challenges." *Autom. Constr.* 132 (4): 103940. <https://doi.org/10.1016/j.autcon.2021.103940>.
- Papineni, S. 2002. "Blue; A method for automatic evaluation of machine translation." In *Proc., ACL 2002*, 311–318. Stroudsburg, PA: Association for Computational Linguistics.
- Pi, Y. L., N. D. Nath, and A. H. Behzadan. 2021. "Detection and semantic segmentation of disaster damage in UAV footage." *J. Comput. Civ. Eng.* 35 (2): 04020063. [https://doi.org/10.1061/\(ASCE\)Cp.1943-5487.0000947](https://doi.org/10.1061/(ASCE)Cp.1943-5487.0000947).
- Poh, C. Q. X., C. U. Ubeynarayana, and Y. M. Goh. 2018. "Safety leading indicators for construction sites: A machine learning approach." *Autom. Constr.* 93 (Mar): 375–386. <https://doi.org/10.1016/j.autcon.2018.03.022>.
- Pour Rahimian, F., S. Seyedzadeh, S. Oliver, S. Rodriguez, and N. Dawood. 2020. "On-demand monitoring of construction projects through a game-like hybrid application of BIM and machine learning." *Autom. Constr.* 110 (Feb): 14. <https://doi.org/10.1016/j.autcon.2019.103012>.
- Rahman, A., Z. Y. Wu, and R. Kalfarisi. 2021. "Semantic deep learning integrated with RGB feature-Based rule optimization for facility surface corrosion detection and evaluation." *J. Comput. Civ. Eng.* 35 (6): 04021018. [https://doi.org/10.1061/\(ASCE\)Cp.1943-5487.0000982](https://doi.org/10.1061/(ASCE)Cp.1943-5487.0000982).
- Redmon, J., S. Divvala, R. Girshick, and A. Farhadi. 2016. "You only look once: Unified, real-time object detection." In *Proc., Cypri IEEE*, 779–788. New York: IEEE.
- Rennie, S. J., E. Marcheret, Y. Mroueh, J. Ross, and V. Goel. 2016. "Self-critical sequence training for image captioning." In *Proc., IEEE Conf. on Computer Vision and Pattern Recognition*, 7008–7024. New York: IEEE.
- Ryu, J., A. Alwasel, C. T. Haas, and E. Abdel-Rahman. 2020. "Analysis of relationships between body load and training, work methods, and work rate: Overcoming the Novice Mason's risk hump." *J. Constr.*

- Eng. Manage.* 146 (8): 04020097. [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0001889](https://doi.org/10.1061/(ASCE)CO.1943-7862.0001889).
- Simonyan, K., and A. Zisserman. 2014. "Very deep convolutional networks for large-scale image recognition." Preprint, submitted September 4 2014. <https://arxiv.org/abs/1409.1556>.
- Sun, S., C. Luo, and J. Chen. 2017. "A review of natural language processing techniques for opinion mining systems." *Inf. Fusion* 36 (Apr): 10–25. <https://doi.org/10.1016/j.inffus.2016.10.004>.
- Szegedy, C., W. Liu, Y. Jia, P. Sermanet, and A. Rabinovich. 2014. "Going deeper with convolutions." In *Proc., IEEE Conf. on Computer Vision and Pattern Recognition*, 1–9. New York: IEEE.
- Vaswani, A., N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin. 2017. "Attention is all you need." *Adv. Neur. Inf. Process. Syst.* 2017 (1): 30. <https://doi.org/10.48550/arXiv.1706.03762>.
- Wang, C. F., S. E. Antos, and L. M. Triveno. 2021. "Automatic detection of unreinforced masonry buildings from street view images using deep learning-based image segmentation." *Autom. Constr.* 132 (Apr): 103968. <https://doi.org/10.1016/j.autcon.2021.103968>.
- Xiao, B., H. R. Xiao, J. W. Wang, and Y. Chen. 2022. "Vision-based method for tracking workers by integrating deep learning instance segmentation in off-site construction." *Autom. Constr.* 136 (Feb): 104148. <https://doi.org/10.1016/j.autcon.2022.104148>.
- Xiao, B., and Z. H. Zhu. 2018. "Two-dimensional visual tracking in construction scenarios: A comparative study." *J. Comput. Civ. Eng.* 32 (3): 4018006. [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000738](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000738).
- Xu, J., and L. Ding. 2017. "A review of metro construction in China: Organization, market, cost, safety, and schedule." *Front. Eng. Manage.* 4 (1): 4–19. <https://doi.org/10.15302/j-fem-2017015>.
- Xu, J. Q., and H. S. Yoon. 2019. "Vision-based estimation of excavator manipulator pose for automated grading control." *Autom. Constr.* 98 (4): 122–131. <https://doi.org/10.1016/j.autcon.2018.11.022>.
- Xu, K., J. L. Ba, R. Kiros, K. Cho, A. Courville, R. Salakhutdinov, R. S. Zemel, and Y. Bengio. 2015. "Show, attend and tell: Neural image caption generation with visual attention." *Proc. Mach. Learn. Res.* 37 (4): 2048–2057. <https://doi.org/10.48550/arXiv.1502.03044>.
- Yan, X. Z., H. Li, C. Wang, J. Seo, H. Zhang, and H. W. Wang. 2017. "Development of ergonomic posture recognition technique based on 2D ordinary camera for construction hazard prevention through view-invariant features in 2D skeleton motion." *Adv. Eng. Inf.* 34 (11): 152–163. <https://doi.org/10.1016/j.aei.2017.11.001>.
- Yang, J., Z. K. Shi, and Z. Y. Wu. 2016. "Vision-based action recognition of construction workers using dense trajectories." *Adv. Eng. Inf.* 30 (3): 327–336. <https://doi.org/10.1016/j.aei.2016.04.009>.
- Young, P., A. Lai, M. Hodosh, and J. Hockenmaier. 2014. "From image descriptions to visual denotations: New similarity metrics for semantic inference over event descriptions." *Trans. Assoc. Comput. Ling.* 2 (Dec): 67–78. https://doi.org/10.1162/tacl_a_00166.
- Yu, Y., W. Umer, X. Yang, and M. F. Antwi-Afari. 2021. "Posture-related data collection methods for construction workers: A review." *Autom. Constr.* 124 (Apr): 103538. <https://doi.org/10.1016/j.autcon.2020.103538>.
- Yu, Y. T., H. Li, W. Umer, C. Dong, X. C. Yang, M. Skitmore, and A. Y. L. Wong. 2019. "Automatic biomechanical workload estimation for construction workers by computer vision and smart insoles." *J. Comput. Civ. Eng.* 33 (3): 04019010. [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000827](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000827).
- Zhang, H., X. Z. Yan, and H. Li. 2018. "Ergonomic posture recognition using 3D view-invariant features from single ordinary camera." *Autom. Constr.* 94 (4): 1–10. <https://doi.org/10.1016/j.autcon.2018.05.033>.
- Zhang, M., and S. Ge. 2022. "Vision and trajectory-Based dynamic collision prewarning mechanism for tower cranes." *J. Constr. Eng. Manage.* 148 (7): 04022057. [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0002309](https://doi.org/10.1061/(ASCE)CO.1943-7862.0002309).